



SF Private Bank × Lux Industries

Integration Brief

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SF Private Pay · Multi-Currency Accounts · IBAN Issuance · Card Issuance · Merchant Acquiring

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Abstract

This brief enumerates exactly what Lux Industries Inc. needs from SF Private Bank (“SFPB”) to leverage the SFPB banking stack — SF Private Pay (PSP), multi-currency accounts and IBAN issuance, card issuance, and merchant acquiring — under the Lux × Local Partner Strategic Partnership Agreement (50/50 net partnership revenue above documented costs). It is the technical, commercial, and compliance checklist for the joint kickoff. It is confidential between Lux Industries and SFPB and is not for distribution. It is a working document; it is not a contract and does not modify the underlying partnership agreement.

Reference: OG-SP-2026-SFPB (partnership-agreement reference). *Audience:* SFPB technical + commercial leadership. *Purpose:* kickoff agenda + integration prerequisites.

1 Context

Lux Industries operates a native, regulator-neutral financial backend: bankd ledger, forexd FX engine, brokerd smart order router, treasuryd double-entry ledger + 14 leaf adapters, amld AML/sanctions, @luxfi/bridge cross-chain MPC, lux/kms HSM-backed key custody, lux/mpc threshold signing. Customer surfaces sit on top: lux.financial (marketing + product), app.lux.financial (operational console), per-tenant white-label deploys for downstream partners.

SFPB holds the US chartered-bank + BD + ATS + TA + FinCEN MSB + state money-transmitter + CIRO + CSA + FINTRAC license stack and operates SF Private Pay (Zeekash agent rails, MyATMen ATM network, card issuance, ACH / wire / FedNow, stablecoin on/off-ramp).

The partnership is **50/50 net partnership revenue above documented costs**, per the Lux × Local Partner Strategic Partnership Agreement template already shared. This brief enumerates exactly what we need from SFPB at the technical, commercial, and compliance layers so we can wire the integration and start producing partnership revenue.

2 SF Private Pay (PSP)

What SFPB gives us. A regulated US-side payment service provider product covering card acceptance, alternative payment methods, agent cash rails (Zeekash), ATM rails (MyATMen), card issuance, ACH / wire / FedNow, and stablecoin on/off-ramp.

2.1 Technical requirements

- **API base URLs** — production + sandbox.

- **API documentation** — full OpenAPI / Swagger or equivalent spec covering every endpoint (auth, accounts, customers, payments, payouts, refunds, disputes, reports).
- **Auth model** — OAuth client_credentials, API key + HMAC, mTLS, or signed JWT. Token rotation policy and rate limits.
- **Webhook spec** — per event type (payment.authorized, payment.captured, payment.failed, payout.completed, dispute.opened, chargeback.received, ...) with payload schema, HMAC signature header + algorithm, retry policy, idempotency-key header convention. We will subscribe at https://api.lux.financial/webhooks/sfpay/<event_type> with HMAC verification before our callback handler accepts.
- **Idempotency-key convention** — header name and how SFPB dedupes.
- **Test credentials** — test credit card numbers, test bank routing/account numbers, sandbox failure scenarios (insufficient funds, sanctions hit, KYC mismatch, fraud reject, dispute).
- **Rate limits** — per-key request/minute, burst, and tier.

2.2 Capability matrix — one row per rail we plan to surface

Rail	Dir.	Snd	Prod	T+	Per-tx limit	Fee
Card acceptance (Visa/MC/AMEX/Discover)	in	?	?	?	?	?
Card-not-present (e-com)	in	?	?	?	?	?
Card-present (POS)	in	?	?	?	?	?
ACH debit	in	?	?	?	?	?
ACH credit	out	?	?	?	?	?
FedNow	in/out	?	?	?	?	?
RTP (TCH)	in/out	?	?	?	?	?
Wire (Fedwire)	in/out	?	?	?	?	?
Wire (CHIPS)	out	?	?	?	?	?
SWIFT MT103	in/out	?	?	?	?	?
Interac (Canada)	in/out	?	?	?	?	?
Zeekash agent cash-in	in	?	?	?	?	?
Zeekash agent cash-out	out	?	?	?	?	?
MyATMen deposit	in	?	?	?	?	?
MyATMen withdrawal	out	?	?	?	?	?
Stablecoin (USDC/USDT) on-ramp	in	?	?	?	?	?
Stablecoin off-ramp	out	?	?	?	?	?

2.3 Commercial requirements

- **Fee schedule** for each rail (per-transaction, percentage, monthly minimum).
- **Settlement mechanics** — when funds clear; holdback / reserve requirements; FBO vs direct deposit.
- **Revenue-share** mechanics confirming compatibility with the 50/50 above documented costs structure in the partnership agreement.

- **Reporting feeds** — daily reconciliation (CSV/JSON SFTP or webhook push), per-merchant breakdown, fee detail, dispute aging.
- **Chargeback / dispute liability** — who carries the reserve, what's the partner-side process.

2.4 Compliance requirements

- **Shared responsibility matrix** explicitly stating who performs: KYC, KYB, OFAC sanctions screening, PEP screening, AML transaction monitoring, BSA SAR filing, CTR filing, FBAR, FATCA, Reg E disputes, Reg Z disclosures, PCI-DSS scope ownership.
- **PCI-DSS scope** — does SFPay tokenise on the iframe / hosted fields so we stay out of scope, or do we need full SAQ D?
- **Card-network rules attestation** — Visa CISP, Mastercard SDP, AMEX DSOP compliance evidence for SFPay's acquiring stack.

Lux integration target. treasury/pkg/provider/sfpay/ — new adapter implementing the standard Provider interface (CreateAccount, CreatePayment, GetPayment, ListPayments, GetBalance, GetFXQuote / CreateFXConversion if multi-currency, CreateCounterparty, Capabilities). Env-gated on SFPAY_API_KEY. Webhook ingest at forex or bankd with HMAC verify.

3 Multi-currency accounts + IBAN issuance

What SFPB gives us. Account issuance across the SFPB regulated perimeter — USD accounts with ABA routing + account number (US); CAD accounts with transit + account (Canada); GBP, EUR, and other currencies via partner EMI relationships or correspondent banks; named or virtual IBANs where the partner offers SEPA / Faster Payments access.

3.1 Technical requirements

- **Account types** SFPB can issue: checking, savings, custodial (FBO), virtual sub-account, escrow.
- **Account-issuance API** — create per-customer account, list, freeze, close. Per-customer KYC payload requirements.
- **Multi-currency support** — which currencies natively, which via partner EMI, settlement mechanics for cross-currency transfers.
- **IBAN issuance** — does SFPB issue named GBP / EUR IBANs directly or via a UK / EU EMI partner? If partner: who, what passport, shared-responsibility mapping. Or: do we provide our own EU EMI partner and use SFPB only for US/CA?
- **Sweep / cash-management** — overnight sweep, balance interest if any, target-balance rules.
- **Statement + reporting API** — daily balance feed, transaction detail, year-end 1099 / equivalent.

3.2 Commercial requirements

- **Per-account monthly fee schedule.**

- **Per-transaction fees** for each outbound rail at the account level (separate from PSP merchant acquiring).
- **Interest-bearing accounts** — interest split if any.
- **Minimum balance / reserve** required to maintain account.

3.3 Compliance requirements

- **CDD requirements** per customer type (US natural person, US legal entity, US trust, foreign natural person, foreign legal entity). Document checklist + acceptable ID types.
- **OFAC screening cadence** — onboarding + per-transaction + ongoing.
- **CIP/BSA reliance** — can SFPB rely on Simplici.io's KYC/CIP results under a Reliance Agreement (WRA), or does SFPB require independent CDD? Reliance Agreement template?
- **Reg E** dispute handling responsibility.
- **State money-transmitter** licensing footprint — does SFPB cover all 50 states or do we route around unlicensed states?

Lux integration target. `treasury/pkg/provider/sfpb-accounts/` — implements `CreateAccount`, `GetAccount`, `ListAccounts`, `GetBalance`, `CreateCounterparty` end-to-end; routes outbound payment rails through SFPay (above) or directly where SFPB's account API exposes them.

4 Card issuance

What SFPB gives us. BIN sponsorship via SFPB's Visa and/or Mastercard principal membership; issuance of debit, prepaid, and (where applicable) credit cards under our brand or partner-tenant brand, physical and virtual.

4.1 Technical requirements

- **Card-issuing platform** — is SFPB on Marqeta, Lithic, Galileo, FIS, i2c, in-house? Whatever it is, we need API access (read-only at minimum; ideally full create / freeze / replace / cancel).
- **Card product configurations** — debit, prepaid (open-loop, closed-loop), credit, virtual-only, single-use, expense-controlled.
- **Funding model** — direct from cardholder account, FBO pooled, ledger-backed-on-our-side with daily settlement to SFPB.
- **Auth stream** — real-time authorization webhook (sub-200ms decision) with our risk engine (amlid + custom rules) participating in the auth decision. Spec for the auth-request payload + decision response.
- **3DS** — 3DS2 challenge integration, ACS provider, fall-back path.
- **Card-network tokenisation** — Visa Token Service / Mastercard MDES support for Apple Pay / Google Pay / Samsung Pay.
- **Physical card production** — vendor (Perfect Plastic / CPI / G+D Mühlbauer / Tag Systems / Idemia), turnaround, custom artwork approval flow, EMV chip personalisation, contactless.

- **PIN management** — PIN issuance, change, unblock; PIN-Verify service.
- **Dispute / chargeback** — flow into the network dispute API; SFPB-side process for representation + arbitration.

4.2 Commercial requirements

- **Setup fees** — per BIN, per card product, artwork approval.
- **Per-card fees** — issuance, monthly active, transaction (network + processing), 3DS, tokenisation, physical-card production.
- **Interchange revenue share** — the headline economic. Visa/MC posted interchange, less network assessments, less processing fees, less SFPB take; net interchange flows into the 50/50 partnership pool.
- **Cardholder funding** — is there a customer-deposit reserve requirement?

4.3 Compliance requirements

- **KYC for cardholders** — CIP requirements; SFPB-required or our WRA-reliance.
- **CARD Act / Reg Z** disclosure ownership for credit / prepaid.
- **Reg E** dispute ownership.
- **OFAC** screening on cardholder onboarding + ongoing.
- **Network rules** — Visa Operating Regulations, Mastercard Rules attestation; PCI-DSS compliance scope for cardholder data on our side.

Lux integration target. treasury/pkg/provider/sfpb-cards/ implements the card-issuance side of the Provider interface (CreateCard, GetCard, FreezeCard, ReplaceCard, ListTransactionsForCard). Auth-stream webhook handled by bankd with amld risk-gate participation. Card-decline reasons emitted to Lux Network as content-addressed audit events.

5 Merchant acquiring (PSP card acceptance for our partners)

What SFPB gives us. Acquiring services for our downstream merchant partners — onboarding, MID provisioning, settlement, dispute management — across the jurisdictions where SFPB or its acquiring partner holds the requisite acquirer licences.

5.1 Requirements

- **Acquirer jurisdictions** — which Visa / Mastercard regions, plus domestic schemes (AMEX, Discover, Diners, JCB, UnionPay, Interac).
- **Onboarding API** — KYB submission, beneficial-ownership attestation, MCC assignment, monthly volume cap.
- **Underwriting timeline** — SLA from KYB submitted to MID issued.
- **MID provisioning** — per merchant, per MCC; descriptor rules, soft-descriptor support, dynamic-descriptor by transaction.

- **Settlement schedule** — T+N per merchant tier, daily ACH or wire, holdback / reserve mechanics.
- **Dispute API** — chargeback receive, representment submit, evidence upload, arbitration escalation.
- **Reporting** — per-merchant daily + monthly fee detail, gross + net + interchange + assessments + processing.

(Compliance posture inherits from §2 SFPay.)

Lux integration target. The same treasury/pkg/provider/sfpay/ adapter handles the merchant-acquiring surface — same auth, same webhooks, separate endpoint family.

6 Cross-cutting operational + commercial requirements

6.1 Operational

- **Sandbox vs production parity** — every endpoint above must work in sandbox with realistic data, before production credentials issue.
- **API status page** + status webhook.
- **Maintenance windows** — schedule, advance notice, runbook for failover or graceful-degradation.
- **On-call escalation path** — named technical liaison, on-call rotation contact, SLA on critical incidents (P0 / P1 / P2 / P3 with response + resolution targets).
- **Production cutover plan** — phased rollout (shadow → 1% → 10% → 100%), per-product enablement, named go/no-go gates.

6.2 Commercial

- **Documented Costs definition** alignment with the partnership agreement (which SFPB-incurred costs count as Documented; e.g. network interchange, processing, sponsor fees yes; SFPB overhead / marketing / cost of capital no).
- **Net Partnership Revenue reconciliation cadence** — monthly statements within 15 business days of month-end, joint reconciliation meeting within 5 business days of statement delivery, net wire within 30 days.
- **Audit rights** — biannual, on 30 days' notice, at audited party's expense if >5% underpayment.
- **Customer markup retention** — SFPB confirms 100% retention of any customer-facing markups by either party stays with the marking-up party (the standard from the partnership agreement).

6.3 Compliance shared-responsibility matrix

Annex A to the partnership agreement should formalise the SFPB-side vs Lux-side regulated-function allocation across:

- BSA / AML programme ownership

- OFAC sanctions screening
- CIP / CDD / EDD
- Suspicious Activity Reports (FinCEN)
- Currency Transaction Reports (FinCEN)
- FBAR / FATCA / CRS / GoAML
- Reg E (electronic fund transfers) dispute resolution
- Reg Z (truth in lending) disclosures
- Reg D (regulation D, account-class restrictions if any)
- Network operating-rules compliance (Visa CISP, MC SDP)
- PCI-DSS scope ownership per integration surface
- Card Act disclosures (where applicable)
- State money-transmitter licensing footprint
- CIRO / CSA Canadian rules
- FCA / FINTRAC / EBA cross-border rules where SFPB extends

7 What we will give SFPB in return

For reference, the symmetric obligations from the partnership agreement that satisfy SFPB's side of the deal:

- **Simplici.io** white-labelled onboarding, KYC, AML, sanctions, PEP with the per-check pass-through pricing in the agreement; SFPB can rely on Simplicio under a WRA.
- **Lux Network** content-addressed audit substrate covering every state transition relevant to SFPB-supervised activity, with cryptographic commitments retainable for the durations required by CBS, FinCEN, OFAC, FATCA.
- **Identity** — every customer onboarded once, used across every product surface (account, card, FX, crypto) with consistent reference IDs.
- **Distribution** — our customer base served through SFPB-issued accounts, SFPB-issued cards, SFPB-acquired merchants. Concrete volume forecast in the partnership-agreement schedule.
- **Net Partnership Revenue** monthly statements and 50/50 settlement on the agreed cadence.
- **Operational integration** of our risk and AML rules into the SFPB auth stream so SFPB sees consistent risk-gate decisions.

8 Decision checklist for the joint kickoff call

To convert this into a green-lit project we propose the following items as agenda for the kickoff:

1. SFPB confirms API documentation will be shared under NDA within 5 business days.

2. SFPB confirms a sandbox tenant will be provisioned for Lux engineering within 10 business days.
3. SFPB confirms the named technical liaison + on-call escalation path.
4. SFPB confirms the shared-responsibility matrix (Annex A above) is acceptable in principle and identifies any function it would prefer to retain or delegate.
5. SFPB confirms its UK/EU IBAN partner (if any) and the operational model for cross-currency settlement.
6. SFPB confirms the card BIN sponsor / issuing platform and any prior commercial dependencies (e.g. existing Marqeta or Lithic relationship) that we should integrate to vs around.
7. Joint sign-off on the partnership-agreement Annex A (shared-responsibility) and Annex B (Documented Costs definition) within 15 business days.

9 Phased deliverable timeline

Phase	Deliverable	Owner	ETA
0	Mutual confidentiality executed	Both	5 BD
0	Sandbox creds for SFPay + accounts + cards	SFPB	10 BD
0	Annex A + Annex B sign-off	Both	15 BD
1	treasury/pkg/provider/sfpay/ PR open	Lux	4 wks
1	treasury/pkg/provider/sfpb-accounts/ PR open	Lux	4 wks
1	treasury/pkg/provider/sfpb-cards/ PR open	Lux	6 wks
1	Shadow-mode pilot (read-only, no funds)	Joint	8 wks
2	1% live traffic gate	Joint	12 wks
2	10% live traffic gate	Joint	16 wks
3	100% live; first Net Partnership Revenue settlement	Joint	24 wks
3	Audit rights exercised once for baseline	Joint	28 wks

Document prepared by Lux Industries Inc. for SF Private Bank. Confidential and pre-decisional. Reference: OG-SP-2026-SFPB (partnership-agreement reference).